# Net craft project Report

- **Project: Net Craft <Benedict Kam, s5>**

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# Introduction

- In today's interconnected world, network security is paramount to protect sensitive information and maintain the integrity of critical systems. This report delves into the essential tools and techniques used to analyze network configurations, identify potential vulnerabilities, and mitigate security risks. The primary aim of this report is to provide a comprehensive overview of network analysis methodologies and their significance in a cybersecurity context. It examines the use of command-line tools like `ipconfig` and `arp`, along with the powerful network protocol analyzer Wireshark, to gather critical information about network configurations, ARP tables, and network traffic.

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# Methodologies

- We use the command `ipconfig /all` to display the DHCP Server information or other information that are related to the network we are connected to.
    - `ipconfig` - This command is short for "Internet Protocol Configuration," is used to display the network settings of a **WINDOWS** computer. It shows the IP address, subnet mask, and default gateway for all network connections. For MAC users, they will be using `ifconfig` for their terminal.
    - `/all` - This flag instructs the command `ipconfig` to display a comprehensive set of network configuration information, including details about DHCP servers, DNS servers, MAC addresses, and more. This is helpful for in-depth network troubleshooting and analysis.
- We use the command `arp -a` to display the ARP table information

    - `arp` - This command, short for "Address Resolution Protocol," is used to view and manipulate the ARP cache on a device. The ARP cache stores mappings between IP addresses and MAC addresses of devices on the local network.
    - `-a` - This flag instructs the command `arp` to display all the entries in the ARP cache. This shows the IP address and corresponding MAC address of each device recently communicated with on the network. This information can be useful for network troubleshooting, identifying devices on the network, and security analysis.
- Wireshark
    - Wireshark allows us to capture and  inspect the network traffic in real-time or from a saved capture file. Wireshark provides detailed information about individual packets, including source and destination addresses, protocols used, and payload data. This tool is essential for network troubleshooting, security analysis, and understanding network protocols.
        - To focus the analysis and efficiently extract relevant information from the captured network traffic, the following Wireshark filters were utilized:
            - `dns` This filter isolates DNS traffic, allowing for the examination of DNS queries and responses, which can reveal information about the domains and hosts being accessed.
            - `dhcp` This filter displays DHCP traffic, facilitating the analysis of DHCP server interactions and the assignment of IP addresses to devices on the network.
            - `http` This filter captures HTTP traffic, enabling the inspection of web requests and responses, which can reveal information about websites visited, data transmitted, and potential security vulnerabilities in web applications.

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# Discussion

- Network Configuration Analysis
    - The use of `ipconfig /all` allowed for a comprehensive examination of the network configuration. By utilizing the `/all` flag, detailed information about the system's network interfaces, including IP addresses, subnet masks, default gateways, DNS server addresses, and MAC addresses, was obtained. This information is crucial for understanding the system's network environment and identifying potential security vulnerabilities. For instance, the identification of the DHCP server's IP address can be used to assess the security configuration of the DHCP server and its potential susceptibility to attacks such as DHCP spoofing
- ARP Cache Analysis

    - The `arp -a` command provided insights into the Address Resolution Protocol (ARP) cache. This cache stores mappings between IP addresses and MAC addresses, which is essential for communication on a local network. By examining the ARP cache, we can identify the devices that the system has recently communicated with. This information can be valuable for detecting suspicious devices or identifying potential ARP poisoning attacks, where an attacker attempts to associate their MAC address with another device's IP address to intercept network traffic.
- Network Traffic Capture and Analysis
    - Wireshark enabled the capture and analysis of network traffic, providing a granular view of the communication patterns and protocols used on the network. By inspecting individual packets, we can identify potential security threats, such as malicious traffic, unauthorized access attempts, and data exfiltration. This analysis allows for a deeper understanding of network behavior and aids in the identification of vulnerabilities that may be exploited by attackers.
        - Data observation of DNS packet

| | | | | | | |
|---|---|---|---|---|---|---|
| 722 | 17:27:16.989922 | 192.168.50.36 | 192.168.50.1 | DNS | 82 | Standard query 0x6c17 HTTPS msgstore.www.notion.so |
| 723 | 17:27:16.990637 | 192.168.50.1 | 192.168.50.36 | DNS | 114 | Standard query response 0x01e7 A msgstore.www.notion.so A 172.64.148.154 A 104.18.39.102 |

            - The first query is for `msgstore.www.notion.so`, and the client (192.168.50.36) sends a request to the DNS server (192.168.50.1).
            - The server responds with two IP addresses for the domain: `172.64.148.154` and `104.18.39.102`. These are part of Notion's infrastructure.
        - Data observation of DHCP packets

| | | | | | | |
|---|---|---|---|---|---|---|
| 213 | 17:26:54.520719 | 192.168.50.36 | 192.168.50.1 | DHCP | 342 | DHCP Release - Transaction ID 0x2c80264 |
| 306 | 17:27:02.659003 | 0.0.0.0 | 255.255.255.… | DHCP | 342 | DHCP Discover - Transaction ID 0x78106390 |
| 367 | 17:27:06.096841 | 0.0.0.0 | 255.255.255.… | DHCP | 342 | DHCP Discover - Transaction ID 0x78106390 |
| 413 | 17:27:09.671174 | 192.168.50.1 | 255.255.255.… | DHCP | 342 | DHCP Offer - Transaction ID 0x78106390 |
| 414 | 17:27:09.671651 | 0.0.0.0 | 255.255.255.… | DHCP | 352 | DHCP Request - Transaction ID 0x78106390 |
| 416 | 17:27:09.777151 | 192.168.50.1 | 255.255.255.… | DHCP | 345 | DHCP ACK - Transaction ID 0x78106390 |

            - Release: The client (192.168.50.36) releases its current IP address.
            - Discover: The client sends a broadcast (0.0.0.0 → 255.255.255.255) to find a DHCP server.
            - Offer: The server (192.168.50.1) offers an IP address (192.168.50.36).
            - Request: The client requests the offered IP address.
            - ACK: The server acknowledges, finalizing the assignment of the IP.
        - Data Observation of HTTP packet

| | | | | | | |
|---|---|---|---|---|---|---|
| 493 | 17:27:14.755270 | 192.168.50.36 | 103.1.139.88 | HTTP | 165 | GET /connecttest.txt HTTP/1.1 |
| 495 | 17:27:14.759742 | 103.1.139.88 | 192.168.50.36 | HTTP | 241 | HTTP/1.1 200 OK (text/plain) |
| 827 | 17:27:17.315667 | 192.168.50.36 | 157.240.13.55 | HTTP | 59 | POST /chat HTTP/1.1 |
| 5007 | 17:28:34.621109 | 192.168.50.36 | 199.232.46.172 | HTTP | 340 | GET /msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?753ba2620ea0af21 HTTP/1.1 |
| 5009 | 17:28:34.626248 | 199.232.46.172 | 192.168.50.36 | HTTP | 255 | HTTP/1.1 304 Not Modified |

            - A **GET** request for `/connecttest.txt` (likely a connectivity test file) is made to a server at `103.1.139.88`. The server responds with `200 OK`, indicating success.

- Another **POST** request is made to `/chat`, indicating interaction with a chat application (server at `157.240.13.55`).
- Lastly, a **GET** request fetches a trusted certificate update from Microsoft servers (`199.232.46.172`), which returns a `304 Not Modified` status, indicating no new updates.

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# Conclusion

- From a cybersecurity practitioner's perspective, this report's findings emphasize the vital importance of network awareness and proactive monitoring. Our analysis of network configuration data (`ipconfig /all`), ARP tables (`arp -a`), and network traffic (Wireshark) reveals potential vulnerabilities and underscores the need for robust security measures.
  The analysis also exposes several limitations. While identifying DHCP server information is crucial, this knowledge also highlights vulnerability to DHCP spoofing attacks when servers lack adequate protection. Similarly, examining ARP tables helps detect poisoning attempts but cannot prevent them entirely. Wireshark traffic analysis, though powerful for identifying suspicious activities, demands continuous monitoring and expert analysis—resources that many organizations may find difficult to maintain.
  These challenges point to the need for a multi-layered security approach combining proactive measures, continuous monitoring, and user education. Our upcoming recommendations will outline specific actions and strategies to enhance network security and address these identified risks.

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# Recommendations

- Based on the findings and conclusions of this study, the following recommendations are proposed to enhance network security and mitigate the identified risks.
  - Secure DHCP Server Configuration: To mitigate the risk of DHCP spoofing identified through the analysis of DHCP server information (ipconfig /all).
    - Implement DHCP snooping on network switches to prevent unauthorized DHCP servers from operating on the network.
    - Configure IP address reservations to assign fixed IP addresses to critical devices based on their MAC addresses, ensuring that only authorized devices receive IP addresses from the legitimate DHCP server.
  - **Enhance ARP Security:** To address the potential for ARP poisoning highlighted in the ARP cache analysis (`arp -a`)
    - Use static ARP entries for critical devices to prevent ARP cache poisoning, ensuring that the MAC address associated with a given IP address remains constant and cannot be manipulated by an attacker.

- Implement Dynamic ARP Inspection (DAI) on network switches to validate ARP packets and prevent malicious ARP replies, adding an extra layer of protection against ARP poisoning attempts.

- **Improve Network Traffic Analysis:** To enhance the effectiveness of network traffic analysis using Wireshark.
    - Develop a comprehensive incident response plan to guide the actions taken when suspicious activity is detected during network traffic analysis. This plan should include procedures for identifying, containing, and eradicating threats, as well as for recovering from any damage caused.
    - Invest in automated network monitoring tools to complement manual analysis with Wireshark. These tools can help identify anomalies and potential threats more efficiently, allowing security teams to focus on investigating and responding to critical events.

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